

Jordan Banh

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EDUCATION

Carleton University

Bachelor of Engineering, Biomedical and Mechanical Engineering.

Relevant Coursework: Circuits, Control Systems, Signals and Systems, Mechanics of Solids, Dynamics, Engineering Materials

Ottawa, ON

Expected 2028

EXPERIENCE

Maintenance and Reliability Engineering Co-op

Jun 2025 – Aug 2026

CertainTeed Insulation Canada (Saint-Gobain)

Ottawa, ON

- Owned the forehearth guarding design end-to-end: modeled geometry in Fusion 360 and AutoCAD LT, produced RFPs, SOPs, SMPs, and LOTO documentation, and managed contractor and vendor coordination through quoting, demo, installation, testing, and measurement verification.
- Managed the chopper frame replacement project, evaluating a new frame design that cut peak stress 10.5x in the side plane and 6.5x frontally, and leading project management through contractor contact, vendor quoting, and installation.
- Modeled and fabricated a pre-welded burner bushing wall and nozzle retrofit for the plant's fiberizers — eight nozzles per unit — cutting on-machine repair time on a chronic failure point.
- Reselected material for a 24 in curing oven fan operating near 1600°F, comparing mild steel, stainless, and Corten on cost and oxidation resistance.
- Specified and sized IO-Link flow instrumentation for a 6 in compressed air header, evaluating sensor range, accuracy, and process connection against measurement requirements.
- Integrated a TURBO-AIR 2000 centrifugal compressor and its Maestro controller into an Ignition SCADA system to enable real-time monitoring and automated load balancing across plant assets.
- Diagnosed an intermittent star-delta contactor fault on an Atlas Copco ZT160 compressor, isolating the failure through control circuit inspection alongside an electrical contractor.
- Built a machine vision pipeline in Python (OpenCV, homography alignment, consensus OCR) to acquire fault counts directly from equipment HMI screens, replacing manual data transcription.
- Instrumented rotating equipment for vibration measurement, interpreting spectra to identify imbalance and early-stage bearing wear before failure.

Coding and Robotics Instructor

Mar 2024 – Sept 2024

Resolute Center

Ottawa, ON

- Taught Python, C, and LEGO SPIKE sensor and actuator programming to over 60 students aged 7-15.

PROJECTS

Compressed Air System Instrumentation and Control

Sept 2025 – Jun 2026

- Traced excess air production to an uncontrolled decentralized system with no demand measurement; defined the sensing and control architecture to centralize compressor operation.
- Built the business case projecting \$250K in annual electricity savings; the project was accepted and approved for implementation in 2027.

Equipment Data Acquisition and Reliability Tooling

Jan 2026 – Aug 2026

- Built Excel VBA tools computing MTBF and MTTR from breakdown logs, and a browser-based tool for balancing chipper blade sets from measured mass distribution.
- Completed a WCM Yellow Belt machine study on the Strapper lines, documenting failure modes and standard work.

Hed-1: A Robot Alarm Clock (Independent Project, In Progress)

- Designing an owl-shaped robot alarm clock for Human Computer Lab's Mech Challenge with a four-bar linkage and servo neck for head motion; CAD, FEA, and GD&T underway.

Roof Truss Design (Academic)

Apr 2024 – May 2024

- Performed member force analysis to size a roof truss for load distribution; modeled in Fusion 360.

SKILLS

Design: Fusion 360, AutoCAD LT, STEP/CAD geometry editing, stress and fatigue evaluation, material selection, tolerance and fit verification

Hardware: Sensor selection and specification (IO-Link), industrial controls and SCADA (Ignition), motor starter and control circuit troubleshooting, vibration instrumentation, P&ID development

Software: Python (OpenCV, NumPy, openpyxl), C, VBA, JavaScript/React, SAP PM